

PROJECT REQUIREMENTS DOCUMENT (PRD) · ACADEMIC SUBMISSION
BASELINE

Autonomous WhatsApp Decision Infrastructure for Engineering Aspirants

A counseling agent for the ~13 lakh students annually excluded from JoSAA seats, built with structural operational disinterest so that it can advise a family against spending ₹20 lakh.

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01 The Macro Admission Funnel

Every June, approximately thirteen to fifteen lakh Indian engineering aspirants face a high-consequence financial commitment within an intense four-to-six-week counseling window, advised almost exclusively by agents with an explicit commercial interest in their enrollment.

~14.5L

JEE Main Registrations

67,323

Total JoSAA Public Seats

~95%

Excluded from JoSAA (~13L)

20–30%

Dropper Rank Improvement

The Joint Seat Allocation Authority (JoSAA) allocates seats across 23 IITs, 31 NITs, 26 IIITs, and 56 GFTIs. The structural gap between candidate volume and subsidized public capacity produces an addressable cohort of over 1.3 million unplaced candidates annually. For this demographic, private technical education represents an opaque and predatory marketplace.

02 The Eight New-Age Tech Colleges We Track

Over the past four years, an emerging tier of venture-backed software training institutes has entered the undergraduate marketplace. These institutes market modern software pedagogy and placement claims, charging tuition comparable to or exceeding premier private universities.

COLLEGE NAME	EST.	FEES (4 YEAR TOTAL)	BATCH	AUDITED OUTCOME HISTORY	DEGREE AWARDING VEHICLE
Newton School (ADYPU)	2024	₹17.5L – ₹26.9L	~100	Early signals; internships only	Ajeenkya DY Patil University (ADYPU)
Newton School (Rishihood)	2023	₹17.5L – ₹26.9L	~120	Internships verified; full batch graduating	Rishihood University

COLLEGE NAME	EST.	FEES (4 YEAR TOTAL)	BATCH	AUDITED OUTCOME HISTORY	DEGREE AWARDING VEHICLE
Scaler School of Tech (SST)	2023	₹17.0L – ₹24.5L	~60–80	96.3% verified tech internships	Separate partner HEI (Verify accreditation)
Vedam	2025	₹18.0L	~60–80	None (Inaugural cohorts)	ADYPU
Intellipaath IST	2025	₹16.0L	63	None (24 unpaid training internships)	S-VYASA (Deemed-to-be University)
NIAT	2023	₹8.0L – ₹18.0L	~80	8.5 LPA reported avg / 24 LPA high	ADYPU
Veloces	2024	₹10.0L – ₹16.0L	–	None (Zero graduating batches)	ADYPU
Alta School of Tech	2026	₹9.8L – ₹15.0L	–	None (Newest entrant)	ADYPU

“Five of the eight institutions possess zero graduating outcome history, yet command ₹16 to ₹27 lakh. The category’s central question is whether a family should bet a fifth of its net worth on an unproven entity.”

03 Competitive Landscape & Market Differentiation

ADVISORY PLATFORM	COMMERCIAL MODEL	PRIMARY STRUCTURAL FLAW	IMPACT ON STUDENT
Aggregator Portals (Shiksha, Collegedunia)	Cost-per-Lead & Referral fees (₹20,000–₹60,000 / seat)	Ranks colleges by commercial commission rather than student fit.	Student is pushed to private partners paying the highest referral fees.
YouTube Reviewers & Influencers	Brand sponsorships & sponsored campus tours	Highlight top 1% package outliers while concealing dropouts and debt burden.	Creates unrealistic salary expectations and downplays loan traps.
Coaching Class Counselors	Repeat drop-year batch quota targets	Advise another drop year even when rank improvement probability is <20%.	Burn a year of student life to collect coaching tuition fees.
Campus Critique Bot (This Project)	Independent Trust & ₹99 Senior Calls	Zero referral commission. The only party capable of advising against spending!	Protects family net worth, stress-tests loan math, and equips student to audit sales reps.

04 Product Vision & Locked Decision D1

LOCKED DECISION D1 • STRUCTURAL OPERATIONAL DISINTEREST

The bot’s primary performance metric is never admissions. The platform accepts zero commercial referral revenue from any college. The system must possess the structural capability and conversational duty to state:

“Based on your family’s annual income and risk parameters, you should not spend ₹18–24 lakh on any institution in this category. Here is what I would do instead.”

The operational metric `advised_against_spending_rate` must remain strictly greater than zero (target: 10%–20%). A 0% rate over any 30-day window is classified as a

05 User Personas & Psychological Archetypes

PERSONA	DEMOGRAPHICS & SCORES	PSYCHOLOGICAL REALITY	BOT OBJECTIVE
Aarav (The Panic-Stricken Aspirant)	18 yrs, Lucknow, 12th: 84%, JEE Main: 78.4%ile (~3.13L rank)	Emotional collapse. Believes "life is over". Overwhelmed by contradictory advice. Father willing to take ₹20L loan.	De-escalate panic, compute realistic rank, burst the binary trap, execute loan stress-test, talk down from expensive pick.
Ramesh (The Risk-Conscious Parent)	49 yrs, Tier-2 Govt Employee, Income ~₹6L/yr	Willing to sacrifice savings for son's tech career; dreads fraudulent degrees and property confiscation.	Provide exact EMI math, unvarnished collateral rules, and UGC partner accreditation facts.
Priya (The Conflicted Dropper)	19 yrs, Dropper, improved from 62%ile to 81%ile	Deep fatigue. Parents refuse second drop. Wondering if new-age tech colleges offer genuine acceleration.	Deliver honest statistical odds of second drops (<5% positive migration), evaluate curriculum depth, supply audit questions.

06 The 7-Stage Conversational Model

Rather than deploying a rigid decision tree or an unconstrained chatbot, the system executes an empathetic 7-stage state machine modeled after an authoritative, grounded college senior.

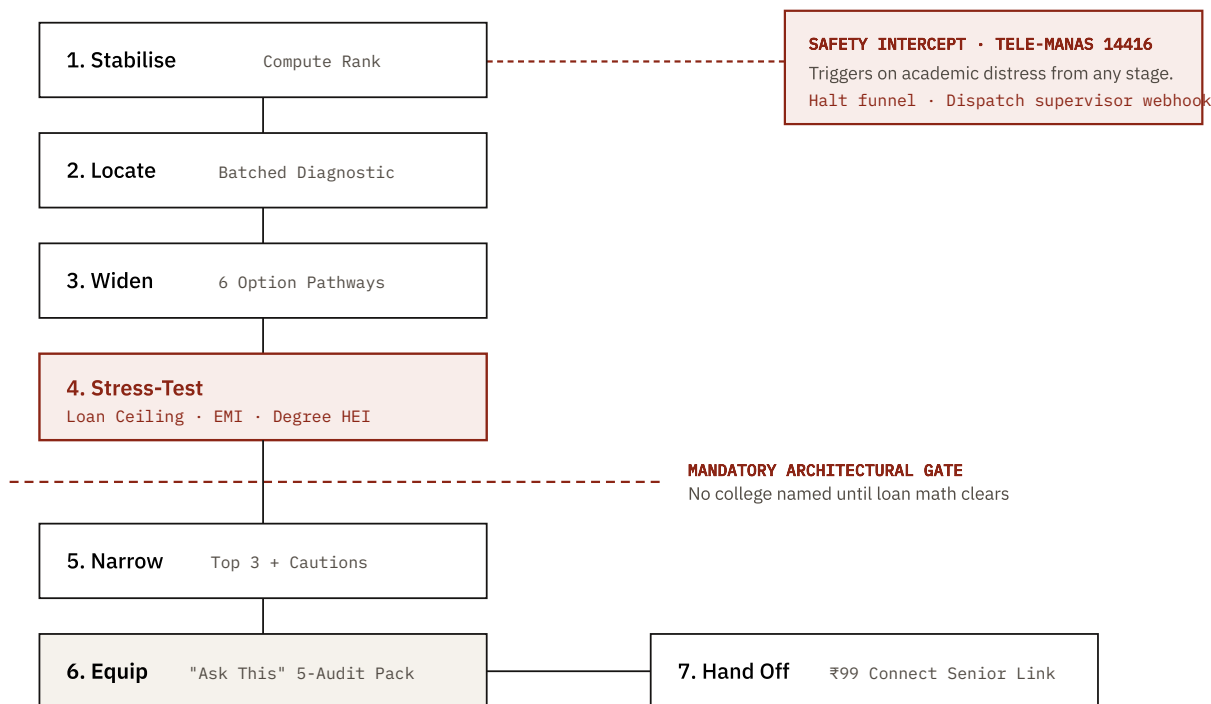


FIGURE 1 The 7-Stage State Machine with Mandatory Gate and Psychological Distress Divergence.

07 MoSCoW Functional Requirements Matrix

ID	PRIORITY	FEATURE DESCRIPTION	ACCEPTANCE CRITERIA
REQ-M01	Must Have	Deterministic Rank Estimation	Given a percentile $P\%$, computes indicative rank using formula $R = (100 - P) \times 14500$. Appends NTA shift normalisation disclaimer.
REQ-M02	Must Have	Loan Capacity Calculation	Caches family income; calculates $4 \times$ borrowing ceiling; flags property collateral requirement if loan $> ₹7.5$ lakh.

ID	PRIORITY	FEATURE DESCRIPTION	ACCEPTANCE CRITERIA
REQ-M03	Must Have	The Mandatory Financial Gate	Permanently blocks college name emission until all three stress-test dimensions (Loan, Downside, Degree) are completed and cleared.
REQ-M04	Must Have	Mandatory Institutional Cautions	Outbound college recommendations must contain the exact audited caution string from <code>wa_college_facts</code> . Rail 4 blocks output if omitted.
REQ-M05	Must Have	Hinglish Distress Safety Intercept	5-boolean classifier intercepts guilt framing; halts funnel within 300ms; dispatches Tele-MANAS 14416 crisis helpline and supervisor webhook.
REQ-S01	Should Have	Unlisted College Demand Tracking	Disclaims lack of verified data when candidate asks about non-indexed colleges; records query in <code>wa_college_demand</code> .
REQ-S02	Should Have	4000ms Ingestion Debouncing	QStash sliding buffer aggregates successive messages sent within 4 seconds into a single unified execution prompt.
REQ-C01	Could Have	Longitudinal Review Flywheel	Automated consent-based WhatsApp pings at 3, 6, and 12 months post-enrollment to collect structured reviews.
REQ-W01	Won't Have (v1)	Predictive AI Cutoff Generator	Strictly banned. No model will ever estimate state counseling cutoffs without verified official data.

08 Messaging UI/UX & WhatsApp Rules

Linguistic Register & Persona Guidelines

The bot converses in an authentic, respectful **Hinglish senior register** (an experienced college senior or older sibling who has successfully navigated counseling). Explicit rules:

- **Banned Superficial Cheer:** Never use *"Don't worry, everything happens for a reason!"* Instead use: *"78.4 percentile after an entire year of preparation genuinely stings. I am not going to sugarcoat that."*
- **Banned Promotional Vocabulary:** Automated regex filters block: *"guaranteed placement", "100% placement", "limited seats", "hurry up", "instant admission"*.
- **Message Length Constraints:** Every outbound bubble is constrained to **under 140 words** to prevent cognitive fatigue.

09

Transcript-Derived KPI Framework (Decision D4)

100% of performance metrics are derived programmatically from the conversational transcript without intrusive surveys.

Metric Name	Derivation Methodology	Target	Strategic Purpose
advised_against_spending_rate ★	Conversations where Stage 4 resulted in negative spending recommendation	10%–20% (>0%)	Proof of operational independence. 0% for a month is a P0 bug.
stage_depth_median	Deepest stage reached across all sessions (1 to 7)	≥ 5.0	Validates conversational engagement and low drop-off.
stress_test_completion_rate	Conversations completing all three tests (Loan, Downside, Degree)	≥ 60%	Measures adherence to the financial gate.
option_set_expansion_rate	Candidates with ≥1 new option pathway considered at close vs intake	≥ 70%	Proves the bot burst narrow binary thinking.
dont_know_rate	Queries where the bot disclaimed lack of verified primary data	> 0%	Measures resistance to hallucinating unindexed colleges.

METRIC NAME	DERIVATION METHODOLOGY	TARGET	STRATEGIC PURPOSE
ungrounded_numeric_claims	Numbers in output missing from retrieved Fact Block	0 (Zero)	Absolute protection against fabricated fees or salaries.
distress_routing_recall	Academic crisis triggers routed to Tele-MANAS	100% (Zero Error)	Safety invariant for vulnerable adolescents.

10 Ethics, Safety & Tele-MANAS (14416)

DPDP Act 2023 Compliance for Minors

- **Isolated Data Silo (Decision D9):** Minors' phone numbers, family income, and session data live in a dedicated Supabase schema (`wa`) completely separated from web application databases.
- **No National Identifiers:** The system strictly prohibits collecting Aadhaar numbers, application passwords, or banking credentials.
- **Session Purge Policies:** Session transcripts are automatically anonymized following the close of each admission cycle.

Mental Health Crisis Protocol: Tele-MANAS (14416)

The system integrates **Tele-MANAS (14416 / 1800-891-4416)**, the national 24/7 mental health counseling helpline launched by the Ministry of Health and Family Welfare (MoHFW) with NIMHANS as the nodal agency, spanning 53 tele-mental health operational cells across 36 states and UTs. When acute distress or self-harm ideation is detected, the counseling funnel halts immediately, a supportive response is delivered, and emergency alerts are dispatched to platform administrators.

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Roadmap & Delivery Milestones

PHASE	MILESTONE NAME	DURATION	CORE DELIVERABLES & EXIT CRITERIA
Phase 0	Infrastructure Foundations	1 Wk	Meta Business Account verified; Supabase <code>wa</code> schema migrated; CI pipeline running.
Phase 1	Deterministic Core (Zero LLM)	2 Wks	All 7 stages operational with zero model calls. Buttons, SQL facts, compute, and match engine fully unit-tested.
Phase 2	Model & Guardrail Engine	2 Wks	Vercel AI SDK integration, Hinglish 5-rail guardrails, decomposed safety classifier, numeric sweep rail live.
Phase 3	Evaluation & Red-Teaming	2 Wks	Promptfoo CI test suite (50+ vulnerability scans), self-hosted Langfuse OTel tracing, transcript replay testing.
Phase 4	Private Pilot (50 Students)	2 Wks	Pilot deployment on dedicated test number; transcript calibration; unit economic audit.
Phase 5	Production Launch & Flywheel	Ongoing	Public deployment; review collection triggers for July-enrolled students; continuous optimization prior to June 2027 peak.

12

Faculty Traceability & Implementation Boundary

STATUS CLARIFICATION

This PRD is the product specification and evaluation baseline. It does not claim that Meta credentials, the WhatsApp Business Account, Supabase production, or the private pilot are already live.

This page is the browser version of the downloadable [PRD PDF](#). Implementation contracts are covered in the [TRD](#); behavioural evidence is in [conversation.html](#).

REQUIREMENT	ARCHITECTURAL CONTROL	VALIDATION ARTEFACT
REQ-M01 · rank estimate	Layer A deterministic compute	Rank unit tests and Aarav replay
REQ-M02 · loan/EMI	Pure arithmetic with dated assumptions	Formula tests and Stage 4 replay
REQ-M03 · financial gate	FSM invariant before Stage 5	Premature-recommendation red team
REQ-M04 · caution	Fact-block caution required by output rail	Grounding and output scanner tests
REQ-M05 · distress	Rail 1 pre-emption and supervisor alert	Hinglish crisis Promptfoo suite
REQ-S01 · unknown college	<code>wa_college_demand</code> event	VIT/unlisted-college scenario
REQ-S02 · debounce	QStash sliding buffer and idempotency	Burst-message integration test

Product invariants

1. Crisis signals stop the funnel immediately.
2. College recommendations require loan, downside and degree checks.
3. Every number is computed, looked up from a versioned fact, or omitted.
4. Every recommendation includes a caution; “none are safe” is valid.
5. Students can reply `STOP` to end follow-up.

Companion Technical Requirements Document available at [trd.html](#). Conversational simulations available at [conversation.html](#).